

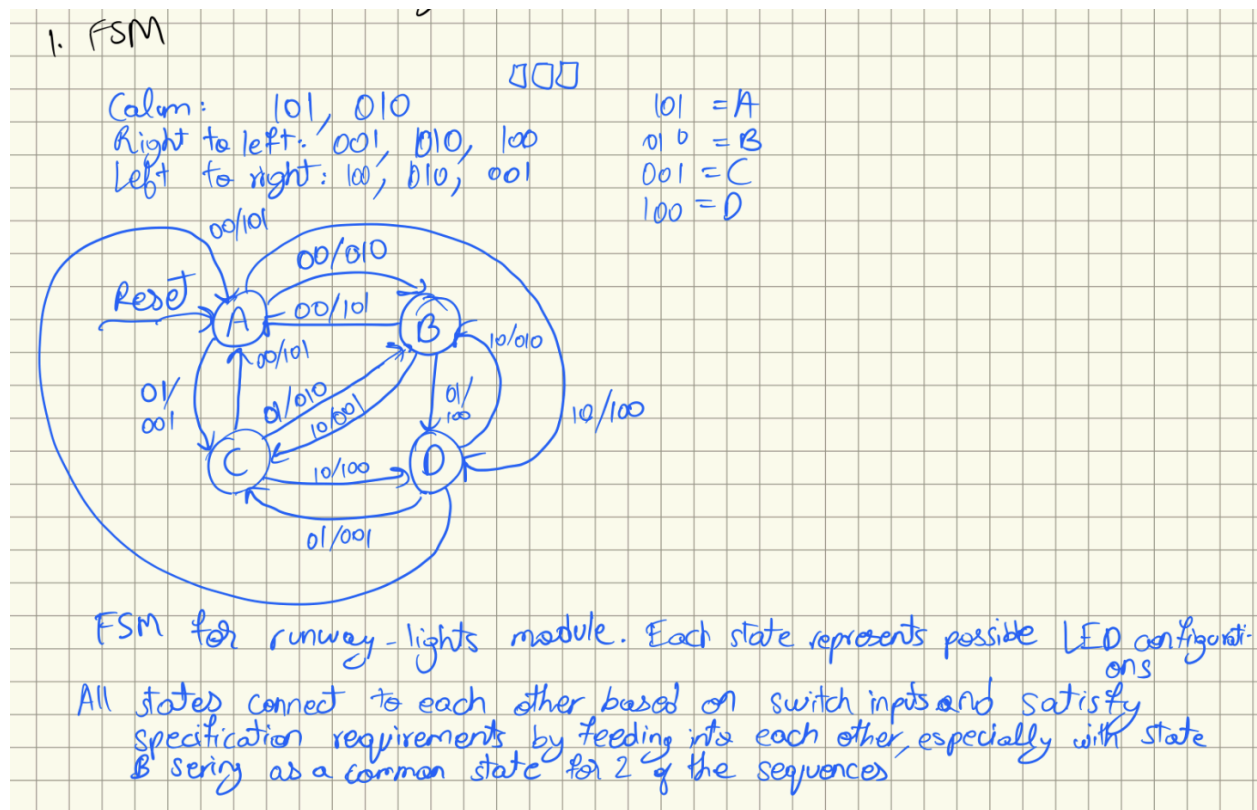
Header



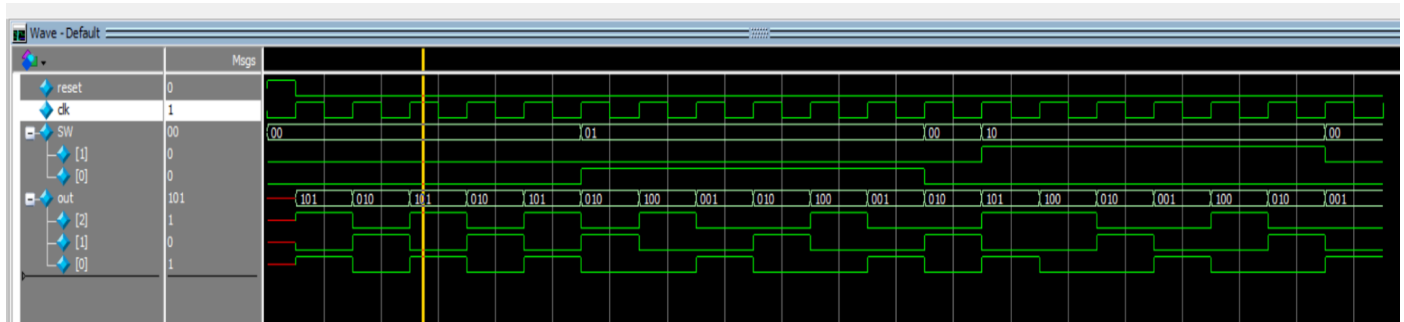
Lab Number: 5

Lab Name: Sequential Logic

Q1.



Q2.



This screenshot shows the entire modelsim screenshot – the results from the testbench for the runway lights module. As we can see there is a clk signal from which all the other signals are being affected. We can see that the switch inputs occur over longer times – this shows that the output signals are being affected according to the clock as well. The output signals correctly map to the desired behavior. This is shown by the yellow line – the cursor and values all over.

We can see that with switch input 00 the output is 101 – the encoding for one of the states during calm conditions. Looking right beside it, we see it flips to 010 – a common transition state – and then goes back 101 – looping through these 2 states in “calm” conditions – which is exactly desired behavior. Similarly, the other two conditions exhibit the correct behavior.

Q3.

Analysis & Synthesis Resource Utilization by Entity										
<<Filter>>										
	Compilation Hierarchy Node	Combinational ALUTs	Dedicated Logic Registers	Block Memory Bits	DSP Blocks	Pins	Virtual Pins	Full Hierarchy Name	Entity Name	Library Name
1	DE1_SoC	30 (0)	29 (0)	0	0	67	0	DE1_SoC	DE1_SoC	work
1	clock_divider:cdiv	26 (26)	26 (26)	0	0	0	0	DE1_SoC clock_divider:cdiv	clock_divider	work
2	runway_lightsFSM:runway_lights	4 (4)	3 (3)	0	0	0	0	DE1_SoC runway_lightsFSM:runway_lights	runway_lightsFSM	work

Time Spent: 3.5 hours